

The Perron vector of a totient Gram matrix and its harmonic approximation

Jeffery Kline

Version 0.1.0 (release candidate), September 2026

Abstract

Take the $n \times n$ matrix R_n whose (j, k) entry counts the integers up to j/k that are coprime to j . Its first column holds Euler's totient $\varphi(1), \dots, \varphi(n)$, and we call its Gram matrix $Q_n = R_n^\top R_n$ a totient Gram matrix. Every entry of Q_n is positive, so Q_n has a unique positive unit eigenvector for its largest eigenvalue, the Perron vector v_n . The author's 2020 paper, which introduced these matrices, showed that Q_n has determinant one, with the Möbius vector as the first column of Q_n^{-1} , and observed numerically that v_n is nearly a multiple of the harmonic vector $h_n = (1, 1/2, \dots, 1/n)$. It conjectured that

$$\lim_{n \rightarrow \infty} n \|h_n - (v_n^\top h_n) v_n\|_\infty = 1,$$

that is, that the part of h_n left after projecting onto v_n has sup norm exactly $1/n$ to first order. We prove this. We also identify the residual: multiplied by n , its k th coordinate tends to $U(t)$ along $k/n \rightarrow 1/t$, for an explicit arithmetic function U , and the conjecture is equivalent to the inequality $|U(t)| \leq 1$ for $t \geq 1$. That inequality holds by an exact integer computation for $t \leq 10^6$ and, for larger t , by an analytic bound resting on Ramaré's explicit estimate for the harmonic Möbius sum. The same analysis improves two results of the 2020 paper. Its Proposition 8, $(c + o(1))n^3 \leq \Lambda_n \leq (\pi^2/18)n^3 + O(n^2)$ for the largest eigenvalue, with $c = \frac{1}{3} \prod_p (1 - 2/p^2 + 1/p^3) \approx 0.1427$, becomes $\Lambda_n \sim c\zeta(2)n^3$. Its Corollary 9, $|v_n^\top \mu_n| \leq (c^{-1/2} + o(1))n^{-3/2}$, becomes $v_n^\top \mu_n \sim n^{-3}/(c\zeta(2)^{3/2})$. Finally, since $\sum_{k \leq n} \mu(k)/k = h_n^\top \mu_n$, one might estimate that sum by replacing h_n with its projection onto v_n . An exact identity shows the replacement returns the sum unchanged up to $O(\log n/n)$; no new cancellation follows.

1 Introduction

Take the $n \times n$ matrix R_n whose (j, k) entry counts the integers up to j/k that are coprime to j :

$$R_n(j, k) = \#\{1 \leq a \leq \lfloor j/k \rfloor : \gcd(a, j) = 1\}.$$

Its first column holds Euler's totient $\varphi(1), \dots, \varphi(n)$ and its diagonal is all ones. Its Gram matrix $Q_n = R_n^\top R_n$ has positive entries, so Q_n has a simple largest eigenvalue Λ_n and a unique positive unit eigenvector v_n , its Perron vector. We call Q_n a totient Gram matrix.

These matrices come from [1], where $R_n = D_n^{-1} T_n D_n$ for the divisibility matrix $D_n(i, j) = 1_{j|i}$ and the lower-triangular all-ones matrix T_n . That paper proves

$$R_n \mu_n = e_1, \quad Q_n \mu_n = e_1,$$

for the Möbius vector $\mu_n = (\mu(1), \dots, \mu(n))^\top$, so Q_n has determinant one and the Möbius vector is the first column of Q_n^{-1} . Its Proposition 8 proves $(c + o(1))n^3 \leq \Lambda_n \leq (\pi^2/18)n^3 + O(n^2)$ with

$c = \frac{1}{3} \prod_p (1 - 2/p^2 + 1/p^3) \approx 0.1427$, and its Corollary 9 proves $|v_n^\top \mu_n| \leq (c^{-1/2} + o(1))n^{-3/2}$, where $c^{-1/2} \approx 2.6467$. It notes numerically that v_n is nearly a multiple of the harmonic vector $h_n = (1, 1/2, \dots, 1/n)^\top$, and closes with the conjecture

$$\lim_{n \rightarrow \infty} n \|h_n - (v_n^\top h_n) v_n\|_\infty = 1. \quad (1)$$

The vector inside the norm is what is left of h_n after projecting it onto v_n . The conjecture says this residual has sup norm exactly $1/n$ to first order.

Main result

Theorem 3.1 proves (1) and describes the residual coordinate by coordinate. Write $r_n = h_n - (v_n^\top h_n) v_n$, $g(a) = \prod_{p|a} p/(p+1)$, $G = \prod_p (1 - 1/(p(p+1)))$, $\kappa = 3/(2G)$, and, for real $t \geq 1$,

$$U(t) = t - \kappa \sum_{a \leq t} g(a) \left(1 - \frac{a^2}{t^2}\right).$$

Then $n r_n(k) \rightarrow U(t)$ whenever $k/n \rightarrow 1/t$, and for each fixed k , $n r_n(k) \rightarrow \rho/k$ for an explicit constant ρ with $|\rho| < 6/\pi^2$. Since $U(1) = 1$, the conjecture is the same as the inequality $|U(t)| \leq 1$ for all $t \geq 1$. The lower bound in (1) is immediate: the last column of R_n is e_n , so $n r_n(n) \rightarrow 1$.

Method

The proof has three parts.

1. *Matrix* (Lemma 3.2). Inclusion–exclusion over the divisors of j gives $R_n(j, k) = \varphi(j)/k + O(\tau(j))$, where τ counts divisors. So R_n is the rank-one matrix $\varphi_n h_n^\top$ plus an error E_n whose Frobenius norm is $O(n \log^{1/2} n)$, against order $n^{3/2}$ for the main term. Perturbation of a rank-one matrix expresses v_n , Λ_n , and r_n through the correction vector $q = E_n^\top \varphi_n / \|\varphi_n\|_2^2$, with Euclidean error $O(n^{-3/2} \log^3 n)$. This part is elementary and uniform in the coordinate.
2. *Arithmetic* (Lemma 3.3). One has $n q_k = -U(n/k) + O(\log^2 n/k)$ uniformly in k , and for $k \leq \log^B n$ one has $n q_k = O(\log^{-A} n)$ for every A . Here the prime number theorem enters, in the form $\sum_{k \leq x} \mu(k) = O(x \exp(-a\sqrt{\log x}))$; nothing unproved is used. The same input gives $U(t) \rightarrow 0$ and $\int_1^\infty U(t) dt/t = 1/(4c) - 1$, where $c = G/(3\zeta(2))$.
3. *Scalar* (Lemma 3.4). On each interval $[m, m+1]$ the function U is convex and given by a closed formula in two partial sums. A program in exact integer arithmetic, with every real constant replaced by an outward-rounded rational interval, checks $U(m) < 3/4$ at every integer $2 \leq m \leq 10^6$, which by convexity and $U(1) = 1$ bounds U above by 1 on $[1, 2]$ and by $3/4$ on $[2, 10^6]$, and checks $U > -3/4$ on each half-unit interval $[m, m + \frac{1}{2}]$ and $[m + \frac{1}{2}, m+1]$ up to 10^6 , using that $t - \kappa A_m$ increases and $\kappa B_m/t^2$ decreases across the interval, where A_m and B_m are the two partial sums. For $t \geq 10^6$, Ramaré’s bound $|\sum_{k \leq x} \mu(k)/k| \leq 1/(12 \log x)$ for $x \geq 687$ [2] gives $|U(t)| \leq 202847/300000 < 0.68$. Together these cover every real $t \geq 1$.

Spectral consequences

The matrix part alone improves two results of [1]. Proposition 4.1 replaces the two-sided bound of Proposition 8 there by the asymptotic $\Lambda_n \sim c\zeta(2)n^3 = (G/3)n^3 \approx 0.2348n^3$, and combines the exact identity $\Lambda_n v_n^\top \mu_n = v_n(1)$ with the eigenvector asymptotics to replace the upper bound $(c^{-1/2} + o(1))n^{-3/2}$ of Corollary 9 there by the asymptotic $v_n^\top \mu_n \sim n^{-3}/(c\zeta(2)^{3/2}) = 3\sqrt{6}n^{-3}/(\pi G)$. Proposition 4.2 bounds the rest of the spectrum by $O(n^2 \log n)$.

A cancellation attempt that fails

Corollary 9 of [1] came from the decomposition $\sum_{k \leq n} \mu(k)/k = h_n^\top \boldsymbol{\mu}_n = (v_n^\top h_n)(v_n^\top \boldsymbol{\mu}_n) + r_n^\top \boldsymbol{\mu}_n$. The first term is now known to be of order n^{-3} , so one asks whether the residual correlation $r_n^\top \boldsymbol{\mu}_n$ can be estimated another way. Section 6 shows that the residual can be replaced by its scalar profile with summed error $O(\log^3 n/n)$, and that, because the profile has controlled variation, known bounds for the Mertens function control the profile correlation $\sum_{k \leq n} \mu(k)U(n/k)/n$. An exact convolution identity then shows that the profile correlation equals $\sum_{k \leq n} \mu(k)/k$ itself up to $O(\log n/n)$. The approximation is accurate but circular for this purpose. No new estimate for Möbius sums is claimed.

Prior work and scope

The matrices, the identities $R_n \boldsymbol{\mu}_n = e_1$ and $Q_n \boldsymbol{\mu}_n = e_1$, the eigenvalue bounds, Corollary 9, and the conjecture itself are from [1]. The totient-sum asymptotic $\sum_{j \leq n} \varphi(j)^2 \sim cn^3$ is classical. The nearest spectral literature concerns Redheffer's matrix, the $(0, 1)$ matrix with (i, j) entry one exactly when $j = 1$ or $i \mid j$, whose determinant is the Mertens function. Barrett, Forcade and Pollington [5] showed that all but $\lfloor \log_2 n \rfloor + 1$ of its eigenvalues equal one and that its spectral radius is asymptotic to $\sqrt{n}$; Vaughan [6] refined that analysis; and Clément and Steinerberger [7] showed that its dominant singular vector is close to the explicit vector $k \mapsto \sum_{d \mid k} 1/d$. Those results concern a different matrix, with spectral radius of order $\sqrt{n}$ rather than n^3 , and none of them addresses R_n , Q_n , or a sup-norm residual limit. The analytic inputs are an effective Mertens-type bound for $\sum_{k \leq x} \mu(k)$, available for example from Lee and Leong [4], and Ramaré's explicit estimate [2], whose corrigendum [3] does not affect the theorem used. A bounded literature search, recorded in the repository file `audit/PRIOR.ART.md`, found no other resolution of (1) and no earlier determination of the constants in Proposition 4.1. That is a statement about the searched corpus, not a priority claim.

Evidence and AI assistance

The argument, the certificate program, and this exposition were developed with AI assistance under the author's direction. One model first proposed the scalar tail bound; another re-implemented it in exact integer arithmetic. The matrix part, the assembled proof, and the certificate were each examined by separate fresh-context model audits, whose reports are preserved in the repository. Such audits are process evidence. They are not peer review, and they do not certify correctness. The certificate is executable, and its receipt can be regenerated with `code/verify_reproduction.py`; Appendix A lists every companion file.

2 Notation

For $1 \leq j, k \leq n$, put

$$R_n(j, k) = \#\{1 \leq a \leq \lfloor j/k \rfloor : (a, j) = 1\}, \quad Q_n = R_n^\top R_n.$$

Equivalently, $R_n = D_n^{-1} T_n D_n$, where $D_n(i, j) = 1_{j \mid i}$ and $T_n(i, j) = 1_{j \leq i}$. Write Λ_n for the largest eigenvalue of Q_n , and v_n for its positive Euclidean-unit eigenvector. Define

$$h_n(k) = \frac{1}{k}, \quad \boldsymbol{\mu}_n(k) = \mu(k), \quad \alpha_n = v_n^\top h_n, \quad r_n = h_n - \alpha_n v_n.$$

The matrix R_n is lower triangular with diagonal one. Its first column is $(\varphi(j))_{j \leq n}$ and its last column is e_n . All entries of Q_n are positive, so the specified Perron vector is well defined. The exact

identities from [1] are

$$R_n \boldsymbol{\mu}_n = e_1, \quad Q_n \boldsymbol{\mu}_n = e_1. \quad (2)$$

Set

$$c = \frac{1}{3} \prod_p \left(1 - \frac{2}{p^2} + \frac{1}{p^3} \right), \quad G = \prod_p \left(1 - \frac{1}{p(p+1)} \right) = 3c\zeta(2), \quad \kappa = \frac{3}{2G},$$

$$g(a) = \prod_{p|a} \frac{p}{p+1}, \quad U(t) = t - \kappa \sum_{a \leq t} g(a) \left(1 - \frac{a^2}{t^2} \right).$$

Empty products equal one. For $0 < t < 1$, the same definition gives $U(t) = t$. Throughout, $L = \log(2n)$, τ is the divisor function, and $M(x) = \sum_{k \leq x} \mu(k)$, $m(x) = \sum_{k \leq x} \mu(k)/k$ are the Mertens and harmonic Möbius sums.

3 The conjecture and its scalar profile

The following theorem settles the conjecture stated at the end of [1], immediately after its Corollary 9, and describes the residual coordinate by coordinate.

Theorem 3.1. *Let v_n be the positive unit eigenvector of Q_n corresponding to its largest eigenvalue, and let $h_n = (1, 1/2, \dots, 1/n)^\top$. Then*

$$\lim_{n \rightarrow \infty} n \|h_n - (v_n^\top h_n) v_n\|_\infty = 1. \quad (3)$$

Write $r_n = h_n - (v_n^\top h_n) v_n$. Moreover, with

$$I = \int_1^\infty \frac{U(t)}{t} dt = \frac{1}{4c} - 1, \quad J = \int_1^\infty \frac{U(t)^2}{t^2} dt, \quad \rho = \frac{J - I}{\zeta(2)},$$

one has $nr_n(k) \rightarrow \rho/k$ for each fixed $k \geq 1$, where $|\rho| < 1/\zeta(2)$, and, for each fixed $t \geq 1$ and any $k_n/n \rightarrow 1/t$,

$$nr_n(k_n) \rightarrow U(t).$$

The proof has a matrix part, an arithmetic profile calculation, and a scalar inequality. We give their statements and the principal estimates, retaining the exact certificate as a companion computation.

3.1 Uniform matrix approximation

Suppress subscripts and put

$$a_j = \varphi(j), \quad S = \|a\|_2^2, \quad H = \|h\|_2, \quad E = R - ah^\top, \quad b = E^\top a, \quad q = b/S.$$

Lemma 3.2. *There are vectors d, e such that*

$$v/v_1 = h + q + d, \quad \|d\|_2 = O(n^{-3/2} L^3), \quad (4)$$

$$r = r_1 h - q + e, \quad \|e\|_2 = O(n^{-3/2} L^3), \quad (5)$$

$$r_1 = \frac{h^\top q + \|q\|_2^2}{H^2} + o(n^{-1}), \quad r_1 = O(L^{5/2}/n). \quad (6)$$

Also $S = cn^3 + O(n^2 L^2)$ and

$$\Lambda = SH^2 \{1 + O(n^{-1/2} L^{1/2})\}, \quad v = h/H + O_2(n^{-1/2} L^{1/2}).$$

Proof outline with the controlling estimates. The divisor expansion gives $|E(j, k)| \leq \min(\tau(j), j/k)$, whence

$$\|E\|_F = O(nL^{1/2}), \quad \|Eh\|_2 = O(n^{1/2}L^{5/2}), \quad \|b\|_2 = O(n^{5/2}L^{1/2}).$$

For the first bound, sum $\min(\tau(j)^2, j^2/k^2)$ over k and then use $\sum_{j \leq n} j\tau(j) = O(n^2L)$. For the second, use $|(Eh)_j| \leq \tau(j) \sum_{k \leq n} 1/k$ and $\sum_{j \leq n} \tau(j)^2 = O(nL^3)$. Rank-one perturbation of $ah^\top$, using the spectral gap of order SH^2 and a standard eigenvector perturbation bound, gives the displayed first-order spectral estimates; the companion note `proof/A04-uniform-linearization.md` carries the details. In particular $\|Ev\|_2 = O(n^{1/2}L^{5/2})$.

Column one of E is zero. Subtracting h_k times the first coordinate of the eigen-equation from its k th coordinate gives

$$\Lambda v_1 = S\alpha + b^\top v, \quad \Lambda(v_k - v_1 h_k) = \alpha b_k + (E^\top Ev)_k.$$

Thus $\alpha/(\Lambda v_1) = S^{-1}(1 + O(L^{5/2}/n))$, and the second identity gives (4). If $z = q + d$, the exact projection formula is

$$r_1 = \frac{h^\top z + \|z\|_2^2}{H^2 + 2h^\top z + \|z\|_2^2}.$$

Here $h^\top q = O(L^{5/2}/n)$ and $\|q\|_2^2 = O(L/n)$. Expansion gives (5)–(6). The totient-square estimate follows by expanding $\varphi(j)^2/j^2$ as a divisor sum; its squarefree coefficient has prime value $-2/p + 1/p^2$. $\square$

3.2 Arithmetic estimates

Lemma 3.3. *Uniformly for $1 \leq k \leq n$,*

$$nq_k = -U(n/k) + O(L^2/k). \quad (7)$$

There are absolute effective $A_0, a_0 > 0$ such that

$$|b_k(n)|/n^2 \leq A_0 k^5 e^{-a_0 \sqrt{\log(2n)}}. \quad (8)$$

Consequently, for each fixed $A, B > 0$, $\max_{k \leq L^B} |nq_k| = O_{A,B}(L^{-A})$. Finally, for some $a_1 > 0$,

$$U(t) = O(e^{-a_1 \sqrt{\log t}}), \quad \int_1^\infty U(t) \frac{dt}{t} = \frac{1}{4c} - 1. \quad (9)$$

Derivation and arithmetic inputs. For (7), divisor expansion first gives

$$\sum_{j \leq y, (j,a)=1} \varphi(j) = \frac{y^2 g(a)}{2\zeta(2)} + O(y\tau(a) \log(2y)).$$

Interchange the count index in $\sum_{j \leq n} \varphi(j)R(j, k) = \sum_{a \leq n/k} \sum_{ka \leq j \leq n, (a,j)=1} \varphi(j)$, and use the estimate for S .

For (8), the analytic input is the classical effective bound $M(x) = O(xe^{-a\sqrt{\log(2x)}})$; see the source locators in Appendix A. Put $f_k(u) = -\{u/k\}$. Exactly,

$$b_k(n) = \sum_{j \leq n} \varphi(j)(\mu * f_k)(j).$$

Let $\mathcal{S}_k$ be the positive integers supported on primes dividing k , and $\mu_k(u) = \mu(u)1_{(u,k)=1}$. The identity $\mu_k = \mu * 1_{\mathcal{S}_k}$, split at $\sqrt{x}$, and the moments

$$\sum_{s \in \mathcal{S}_k} s^{-1} \leq k, \quad \sum_{s \in \mathcal{S}_k} s^{-1/2} \leq k^2$$

give $M_k(x) \ll k^2 x e^{-a' \sqrt{\log(2x)}}$. For a nonprincipal character χ modulo k , the hyperbola identity for $\mu_k * \chi$ uses only this bound and $|\sum_{u \leq y} \chi(u)| \leq k$. The principal convolution is the identity element, handled separately.

Multiplication by $\varphi(u)/u$ is a further convolution with prime-power coefficients $(1 - \chi(p))/p$ for $p \nmid k$, and zero for $p \mid k$. Their absolute 3/4-moment is bounded by

$$\prod_p \left(1 + \frac{2}{p(p^{3/4} - 1)} \right) < \infty.$$

Another split at $\sqrt{x}$ and partial summation therefore give, uniformly in k, χ, x ,

$$\sum_{u \leq x} \varphi(u)(\mu_k * \chi)(u) \ll k^2 x^2 e^{-a'' \sqrt{\log(2x)}},$$

with crude bound $O(kx^2)$. To retain non-unit residues, write $j = su$ with $s \in \mathcal{S}_k$, $(u, k) = 1$, and expand $\sum_{d|s} \mu(d) f_k((s/d)u)$ in characters of the unit group. There are at most k coefficients, each of magnitude at most k . The range $s \leq \sqrt{n}$ costs at most $k^5 e^{-a_0 \sqrt{\log(2n)}}$ after division by n^2 ; the remaining range costs at most $k^5 n^{-1/4}$. This proves (8), shrinking a_0 . The companion note `proof/A06-uniform-small-coordinates.md` gives the uniform constants and all small-argument details.

For (9), set

$$\eta(d) = \frac{\mu(d)}{\prod_{p|d} (p+1)}, \quad D_0(y) = \frac{2y}{3} - \sum_{a \leq y} (1 - a^2/y^2).$$

Then $g = 1 * \eta$ and

$$U(t) = \kappa \sum_{d \geq 1} \eta(d) D_0(t/d), \quad \eta = (\mu/\text{id}) * \beta, \quad (10)$$

where $\beta(p^r) = 1/[p^r(p+1)]$ is nonnegative, $\sum \beta = \zeta(2)$, and $\sum \beta(d) d^\delta < \infty$ for $0 < \delta < 1$. The kernel is bounded, equals $2y/3$ below one, and is $1/2 + O(1/y)$ above one; its variation on $(0, T)$ is $O(\log(2T))$. Partial summation using the classical bound for $m(x)$, followed by the positive β convolution, proves the decay. Thus this proof uses known prime-number-theorem cancellation.

For $0 < s < 1$, finite summation and integration give

$$\int_0^\infty D_0(y) y^{-s-1} dy = \frac{-2\zeta(s)}{s(s+2)}.$$

Writing $B(s) = \sum \beta(d) d^{-s}$, absolute interchange in (10) yields

$$\int_0^\infty U(t) t^{-s-1} dt = \frac{-2\kappa\zeta(s)B(s)}{s(s+2)\zeta(1+s)}.$$

The decay just proved permits $s \downarrow 0$ by dominated convergence; the result is $\kappa\zeta(2)/2$. Subtracting the integral over $(0, 1)$, which equals one, gives (9). $\square$

3.3 The scalar bound and completion

Lemma 3.4 (Computer-assisted scalar inequality). *For every real $t \geq 1$, $|U(t)| \leq 1$, with equality only at $t = 1$.*

Analytic tail and finite certificate. Let $P(u) = \sum_{d \leq u} \eta(d)$ and $\epsilon(t) = \sup_{u \geq t} |P(u)|$. Bernoulli polynomial expansion of D_0 gives

$$D_0(y) = \frac{1}{2} + \frac{B_2(\{y\})}{y} - \frac{B_3(\{y\})}{3y^2} \quad (y \geq 1).$$

Writing $F(y) = B_2(\{y\}) - B_3(\{y\})/(3y)$, one has $|F| \leq K_0 := 1/6 + \sqrt{3}/108 < 183/1000$. Splitting (10) at $d = t$ gives

$$\frac{U(t)}{\kappa} = -\frac{P(t)}{6} + \frac{2t}{3} \int_t^\infty \frac{P(u)}{u^2} du + \frac{1}{t} \sum_{d \leq t} d\eta(d)F(t/d).$$

Since $|d\eta(d)| \leq 1$,

$$|U(t)| \leq \kappa \{5\epsilon(t)/6 + K_0\}.$$

Ramaré's Theorem 1.2 [2] gives $|m(x)| \log x \leq 1/12$ for $x \geq 687$, hence $\sup_{x \geq 1000} |m(x)| \leq 1/50$. The global bound $|m(x)| \leq 1$ follows from $\sum_{d \leq x} \mu(d)[x/d] = 1$. Using $P(u) = \sum_{e \leq u} \beta(e)m(u/e)$ therefore gives

$$\epsilon(t) \leq \frac{\zeta(2)}{50} + \sqrt{1000/t} E_{1/2}, \quad E_{1/2} = \sum_e \beta(e)\sqrt{e}.$$

The certificate encloses $\kappa < 213/100$ and $E_{1/2} < 4$. With $\zeta(2) < 5/3$, for $t \geq 10^6$ this proves

$$|U(t)| < \frac{213}{100} \left[\frac{5}{6} \left(\frac{1}{30} + \frac{16}{125} \right) + \frac{183}{1000} \right] = \frac{202847}{300000} < 0.68.$$

On $[m, m+1]$, put $A_m = \sum_{a \leq m} g(a)$ and $B_m = \sum_{a \leq m} a^2 g(a)$. Exactly,

$$U(t) = t - \kappa A_m + \kappa B_m/t^2, \quad U''(t) = 6\kappa B_m/t^4 > 0.$$

The integer certificate uses outward intervals with denominator 10^{30} , a smallest-prime-factor sieve for g , and rigorously bounded Euler products. It verifies $U(m) < 3/4$ at all 999,999 integer endpoints $2 \leq m \leq 10^6$; $U(1) = 1$ is exact. Convexity covers the upper bound between endpoints. For each of 1,999,998 half-intervals $[a, b]$, it verifies the lower enclosure $a - \kappa A_m + \kappa B_m/b^2 > -3/4$ with outward rounding. These are integer assertions, not floating-point samples. Appendix A identifies the executable, receipt and reproduction command. The finite checks and analytic tail exhaust all real $t \geq 1$. $\square$

Proof of Theorem 3.1. Let $K_n = \lceil L^3 \rceil$. Equations (7)–(9), with the small-coordinate estimate, imply

$$nh^\top q \longrightarrow -I, \quad n\|q\|_2^2 \longrightarrow J.$$

For the first limit, the small coordinates contribute $o(1)$, the summed error above K_n is $O(L^2/K_n)$, and the main sum is $-\sum_{k \geq K_n} U(n/k)/k$. The decay in (9) supplies an integrable envelope at zero after $x = k/n$. For the second limit, the bounded continuous function $U(1/x)^2$ extends to zero with value zero. Equation (6) now gives $nr_1 \rightarrow \rho$.

The Euler factors show

$$\frac{1}{8} < \frac{40}{3\pi^4} < c \leq \frac{5}{24} < \frac{1}{4}.$$

Hence $0 < I < 1$. Lemma 3.4 gives $0 \leq J < 1$, so $|\rho| < 1/\zeta(2)$. Uniformly in the two exhaustive coordinate ranges,

$$nr_k = \begin{cases} \rho/k + o(1), & k < K_n, \\ U(n/k) + o(1), & k \geq K_n. \end{cases}$$

The upper bound in (3) follows. The last column of R is e_n , so $q_n = \varphi(n)/S - 1/n$ and (5) gives $nr_n \rightarrow 1$, proving the lower bound. The moving-coordinate assertion follows from continuity of U and (7). $\square$

4 Sharpening the earlier spectral results

These consequences require Lemma 3.2, but not the scalar certificate or the full conjecture. Proposition 8 of [1] proves $(c + o(1))n^3 \leq \Lambda_n \leq (\pi^2/18)n^3 + O(n^2)$, and its Corollary 9 proves $|v_n^\top \mu_n| \leq (c^{-1/2} + o(1))n^{-3/2}$. The next proposition replaces both bounds by asymptotic equivalents.

Proposition 4.1 (Exact correlation and sharp asymptotic). *For every $n \geq 1$,*

$$0 < v_n^\top \mu_n = \frac{v_n(1)}{\Lambda_n} \leq \frac{1}{\sum_{j \leq n} \varphi(j)^2}. \quad (11)$$

Moreover,

$$\frac{\Lambda_n}{n^3} \rightarrow \frac{G}{3}, \quad (12)$$

$$v_n^\top \mu_n \sim \frac{1}{c\zeta(2)^{3/2}} n^{-3} = \frac{3\sqrt{6}}{\pi G} n^{-3}, \quad (13)$$

$$\alpha_n(v_n^\top \mu_n) \sim \frac{3}{G} n^{-3}. \quad (14)$$

Proof. Symmetry and (2) give $\Lambda_n v_n^\top \mu_n = v_n^\top Q_n \mu_n = v_n(1) > 0$. The Rayleigh quotient at e_1 gives $\Lambda_n \geq S$, and $v_n(1) \leq 1$, proving (11). Lemma 3.2 gives $v_n(1) \rightarrow 1/\sqrt{\zeta(2)}$, $\alpha_n \rightarrow \sqrt{\zeta(2)}$, and $\Lambda_n \sim c\zeta(2)n^3$. Substitution proves the claims. $\square$

Numerically, the leading constants are approximately

$$G/3 = 0.2348140670, \quad \frac{3\sqrt{6}}{\pi G} = 3.3204859112, \quad \frac{3}{G} = 4.2586886415.$$

Thus (12) replaces the two-sided bound of Proposition 8 in [1], whose lower constant is this same c , by an asymptotic with constant $c\zeta(2)$; and (13) replaces the $n^{-3/2}$ upper bound of Corollary 9 there by a positive asymptotic equivalent of order n^{-3} . Even the finite inequality (11), without the conjecture or perturbation analysis, already gives $O(n^{-3})$.

Proposition 4.2 (Remainder of the spectrum). *Order the eigenvalues as $0 < \lambda_{1,n} \leq \dots \leq \lambda_{n,n} = \Lambda_n$. Then*

$$\sum_{i < n} \lambda_{i,n} = O(n^2 \log(2n)), \quad \lambda_{n-1,n} = O(n^2 \log(2n)).$$

For $n \geq 2$, $\Lambda_n/\lambda_{n-1,n} \gg n/\log(2n)$ as $n \rightarrow \infty$.

Proof. Put $u = h/H$. Since $R(I - uu^\top) = E(I - uu^\top)$,

$$\begin{aligned} \operatorname{tr} Q - \Lambda &\leq \operatorname{tr} Q - u^\top Q u = \|R(I - uu^\top)\|_F^2 \\ &\leq \|E\|_F^2 = O(n^2 L). \end{aligned}$$

All eigenvalues are positive. The gap bound follows from (12). $\square$

Remark 4.3. These statements sharpen Proposition 8 and Corollary 9 of [1]. No comparison with all subsequent literature is asserted, and no asymptotic equivalent for the second eigenvalue is established.

5 What Hölder transfers to the harmonic Möbius sum

Theorem 3.1 and squarefree counting give

$$\|r_n\|_\infty = \frac{1 + o(1)}{n}, \quad \|\mu_n\|_1 = \frac{n}{\zeta(2)} + O(\sqrt{n}).$$

Consequently Hölder's inequality yields

$$|r_n^\top \mu_n| \leq \frac{1}{\zeta(2)} + o(1), \quad |m(n)| \leq \frac{6}{\pi^2} + o(1). \quad (15)$$

The original Corollary 9 alone suffices to dispose of the first term in (16), because $\alpha_n \leq \|h_n\|_2 \leq \sqrt{\zeta(2)}$. Proposition 4.1 sharpens that term to $(3/G + o(1))n^{-3}$ but does not improve the limiting constant in (15). This constant is approximately 0.607927; it is not an improvement on known bounds for $m(n)$ that tend to zero.

6 A cancellation attempt and its exact obstruction

Corollary 9 of [1] bounds $|v_n^\top \mu_n|$ by approximately $2.6467n^{-3/2}(1 + o(1))$. Since h_n is close to the scaled vector $\alpha_n v_n$, it is natural to ask whether rapid decay of the latter vector's Möbius correlation forces rapid decay of

$$m(n) = h_n^\top \mu_n = \sum_{k \leq n} \frac{\mu(k)}{k}.$$

The exact decomposition is

$$m(n) = \alpha_n(v_n^\top \mu_n) + r_n^\top \mu_n. \quad (16)$$

The issue is therefore a signed correlation of the residual, not merely its size. This section quantifies that distinction.

6.1 Summing the profile approximation

Define $F_n(x) = U(n/x)/n$ for $1 \leq x \leq n$ and $\mathcal{T}_n = \sum_{k \leq n} \mu(k)F_n(k)$.

Proposition 6.1. *One has*

$$(1 - r_n(1))m(n) = \mathcal{T}_n + O(L^3/n). \quad (17)$$

For every integer $1 \leq K < n$, set $\delta_K = \max_{K \leq j \leq n} |M(j)|/j$ and $D = 1 + 2\kappa$. For all sufficiently large n ,

$$|m(n)| \leq \frac{K/n + \delta_K \{1 + K/n + D \log(n/K)\} + O(L^3/n)}{1 - r_n(1)}. \quad (18)$$

Here the O term may be taken as a nonnegative bound with an absolute implied constant.

Proof. Equations (5) and (7) give

$$r_n^\top \boldsymbol{\mu}_n = r_n(1)m(n) + \mathcal{T}_n + O(L^3/n).$$

Indeed the coordinate errors sum to $O((L^2/n) \sum_{k \leq n} 1/k)$, and Cauchy–Schwarz gives $|e^\top \boldsymbol{\mu}_n| \leq \sqrt{n} \|e\|_2 = O(L^3/n)$. Combine with (16) and (14) to obtain (17).

Away from integers,

$$U'(t) = 1 - \frac{2\kappa}{t^3} \sum_{a \leq t} a^2 g(a).$$

Since $0 < g(a) \leq 1$ and $\sum_{a \leq t} a^2 \leq t^3$, $|U'(t)| \leq D$. Continuity at integers makes F_n absolutely continuous on its interval, with $|F'_n(x)| \leq D/x^2$. Also $|F_n| \leq 1/n$ and $F_n(n) = 1/n$.

Bound the first K terms of $\mathcal{T}_n$ absolutely by K/n . For the remainder, the exact discrete summation-by-parts identity is

$$\begin{aligned} \sum_{k=K+1}^n \mu(k) F_n(k) &= M(n) F_n(n) - M(K) F_n(K+1) \\ &\quad + \sum_{j=K+1}^{n-1} M(j) \{F_n(j) - F_n(j+1)\}. \end{aligned}$$

Its boundary terms are bounded by $\delta_K(1 + K/n)$. The remaining terms have total at most

$$D\delta_K \sum_{j=K+1}^{n-1} j \int_j^{j+1} x^{-2} dx = D\delta_K \sum_{j=K+1}^{n-1} \frac{1}{j+1} \leq D\delta_K \log(n/K).$$

Finally $r_n(1) = O(1/n)$, so the denominator is positive for large n . $\square$

For fixed $T > 1$, take $K = \lfloor n/T \rfloor$. If $M(x) = o(x)$, (18) gives $\limsup |m(n)| \leq 1/T$, and then $m(n) \rightarrow 0$. A quantitative input of the form $M(x)/x = O(e^{-a\sqrt{\log x}})$ similarly yields $m(n) = O(e^{-b\sqrt{\log n}})$ for some $b > 0$, by taking K of order $ne^{-b_0\sqrt{\log n}}$ with $b_0 > 0$ small. This recovers the type of decay supplied by the input, rather than improving it. The proof of the profile estimates already uses classical cancellation, so this is not an independent proof of the prime number theorem.

6.2 An exact convolution identity

Proposition 6.2 (The profile retains the original sum). *For every positive integer n ,*

$$\mathcal{T}_n = m(n) - \frac{\kappa}{n} \sum_{b \leq n} \eta(b) \left(1 - \frac{b^2}{n^2}\right). \quad (19)$$

In particular,

$$|\mathcal{T}_n - m(n)| \leq \frac{\kappa(1 + \log n)}{n}. \quad (20)$$

Before passing to the scalar profile, the exact matrix correction satisfies

$$q^\top \boldsymbol{\mu}_n = \frac{1}{S} - m(n). \quad (21)$$

Proof. Since $g = 1 * \eta$, Dirichlet convolution gives $\mu * g = \eta$. Expanding the finite definition of U and grouping by $b = ak$ yields

$$\begin{aligned}\mathcal{T}_n &= m(n) - \frac{\kappa}{n} \sum_{ak \leq n} \mu(k)g(a) \left(1 - \frac{a^2 k^2}{n^2}\right) \\ &= m(n) - \frac{\kappa}{n} \sum_{b \leq n} \eta(b)(1 - b^2/n^2).\end{aligned}$$

Now $|\eta(b)| \leq 1/b$, which proves (20). Finally (2) gives $E\boldsymbol{\mu}_n = e_1 - am(n)$; multiplying by $a^\top/S$ and using $a_1 = 1$ proves (21). $\square$

Remark 6.3 (What the attempt establishes). The loss in Hölder is real: unsigned summation destroys cancellation. The matrix approximation is accurate enough to survive signed summation with error $O(L^3/n)$, and its scalar profile has controlled variation. Nevertheless, the resulting main term equals $m(n)$ itself up to $O(\log n/n)$. Thus substituting the present approximation does not supply a new estimate for that main term. This is an obstruction to this particular transfer argument, not an impossibility theorem for spectral methods. A stronger conclusion would require an additional estimate for the signed correlation or another arithmetic mechanism. No such improvement is claimed here.

A Companion proofs, certificate and provenance

All paths are relative to the repository root. The component notes below were frozen before the final examination; status labels inside them are historical, and `audit/campaign/FINAL-ADJUDICATION.md` records their completed review.

Material	Repository file
Matrix estimates	<code>proof/A04-uniform-linearization.md</code>
Profile calculation	<code>proof/A05-arithmetic-roadmap.md</code>
Uniform small coordinates	<code>proof/A06-uniform-small-coordinates.md</code>
Decay and Mellin calculation	<code>proof/A07-profile-decay-and-integral.md</code>
Uniform integration	<code>proof/A08-reduction-theorem.md</code>
Scalar proof	<code>proof/A09-scalar-certificate.md</code>
Spectral consequences	<code>proof/A11-spectral-consequences.md</code>
Cancellation attempt	<code>proof/A12-mobius-correlation-attempt.md</code>
Assembled proof packet	<code>proof/proof-bundle.md</code>
Certificate executable	<code>code/certify_scalar.py</code>
Recorded certificate receipt	<code>results/scalar-certificate.json</code>
Reproduction runner	<code>code/verify_reproduction.py</code>
Examination records	<code>audit/campaign/</code>

The certificate program's SHA-256 is

```
08395c08e2be59a5a4e97d234d8af032
ffa7f19d9d383e77723721ba9d551587
```

The line break is typographic; the digest is one string. The program is frozen by this digest and was not edited for this release. Its docstring names the interpretation note by its historical path `notes/A09-scalar-certificate.md`, now `proof/A09-scalar-certificate.md`; its comment `eta(1000)<=1/50` denotes $\sup_{x \geq 1000} |m(x)| \leq 1/50$; and its comment on the endpoint check writes $U(m) \leq 3/4$ although the assertion it labels is strict. From the repository root, the reproduction command is

`python3 code/verify_reproduction.py`

It checks the frozen inputs against `audit/frozen-artifacts.json` and reruns the certificate in a temporary directory, using only the Python standard library. The computation is part of Lemma 3.4; the PDF alone does not replace its executable evidence.

The effective Mertens estimate used in the arithmetic argument is available, for example, from Lee and Leong [4], Theorem 1.1, equation (9), with logarithmic factors absorbed by decreasing the exponential constant. The scalar tail uses Ramaré [2], Theorem 1.2, printed page 1361. The corrigendum [3] does not change that theorem.

This document makes no claim of human refereeing or proof-assistant verification. The release, archive, and citation status of this version is recorded in the repository’s `README.md` and `ADMISSION.md`.

References

- [1] J. Kline, Unital sums of the Möbius and Mertens functions, *Journal of Integer Sequences* **23** (2020), Article 20.8.1. Original article and source: <https://cs.uwaterloo.ca/journals/JIS/VOL23/Kline/kline4.html>.
- [2] O. Ramaré, Explicit estimates on several summatory functions involving the Moebius function, *Mathematics of Computation* **84** (2015), no. 293, 1359–1387. <https://ramare-olivier.github.io/Maths/mcom2914.pdf>.
- [3] O. Ramaré, Corrigendum to “Explicit estimates on several summatory functions involving the Moebius function”, *Mathematics of Computation* **88** (2019), no. 319, 2383–2388, doi:10.1090/mcom/3449. <https://ramare-olivier.github.io/Maths/mcom3449-corrigendum.pdf>.
- [4] E. S. Lee and N. Leong, *New explicit bounds for Mertens function and the reciprocal of the Riemann zeta-function*, preprint, arXiv:2208.06141v5 (9 September 2026); the theorem and equation numbers cited here are those of version 5. <https://arxiv.org/abs/2208.06141v5>.
- [5] W. W. Barrett, R. W. Forcade and A. D. Pollington, On the spectral radius of a $(0, 1)$ matrix related to Mertens’ function, *Linear Algebra and its Applications* **107** (1988), 151–159.
- [6] R. C. Vaughan, On the eigenvalues of Redheffer’s matrix I, in *Number Theory with an Emphasis on the Markoff Spectrum*, Lecture Notes in Pure and Applied Mathematics **147**, Dekker, 1993, 283–296; II, *Journal of the Australian Mathematical Society, Series A* **60** (1996), 260–273.
- [7] F. Clément and S. Steinerberger, On the largest singular vector of the Redheffer matrix, preprint, arXiv:2502.09489 (2025).